

7 Cystic fibrosis is an inherited disorder caused by a mutation in the gene coding for the CFTR protein.

(a) Explain how this mutation results in reduced gas exchange in the lungs.

(3)

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(b) The number of new cases of cystic fibrosis in the USA in 2015 was 964.

It is predicted that this value will have fallen to 839 by 2025.

(i) Calculate the predicted percentage decrease in the number of new cases from 2015 to 2025.

(1)

Answer %



(ii) Explain why the number of cases could fall.

(3)

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*(c) One complication of cystic fibrosis is lung disease caused by the build-up of bacteria in the airways.

This may be due to the inability of the person to clear mucus from the airways.

People with cystic fibrosis can inhale a salt solution to treat their symptoms. This gives improvements in the clearance of mucus, lung function and quality of life, for a short time.

It has been suggested that pre-treatment with drug A could extend the effects of inhalation of a salt solution.



Devise an investigation to determine if pre-treatment with drug A is more effective than inhalation of a salt solution on its own.

Give reasons for your choice of method.

(6)

(Total for Question 7 = 13 marks)

